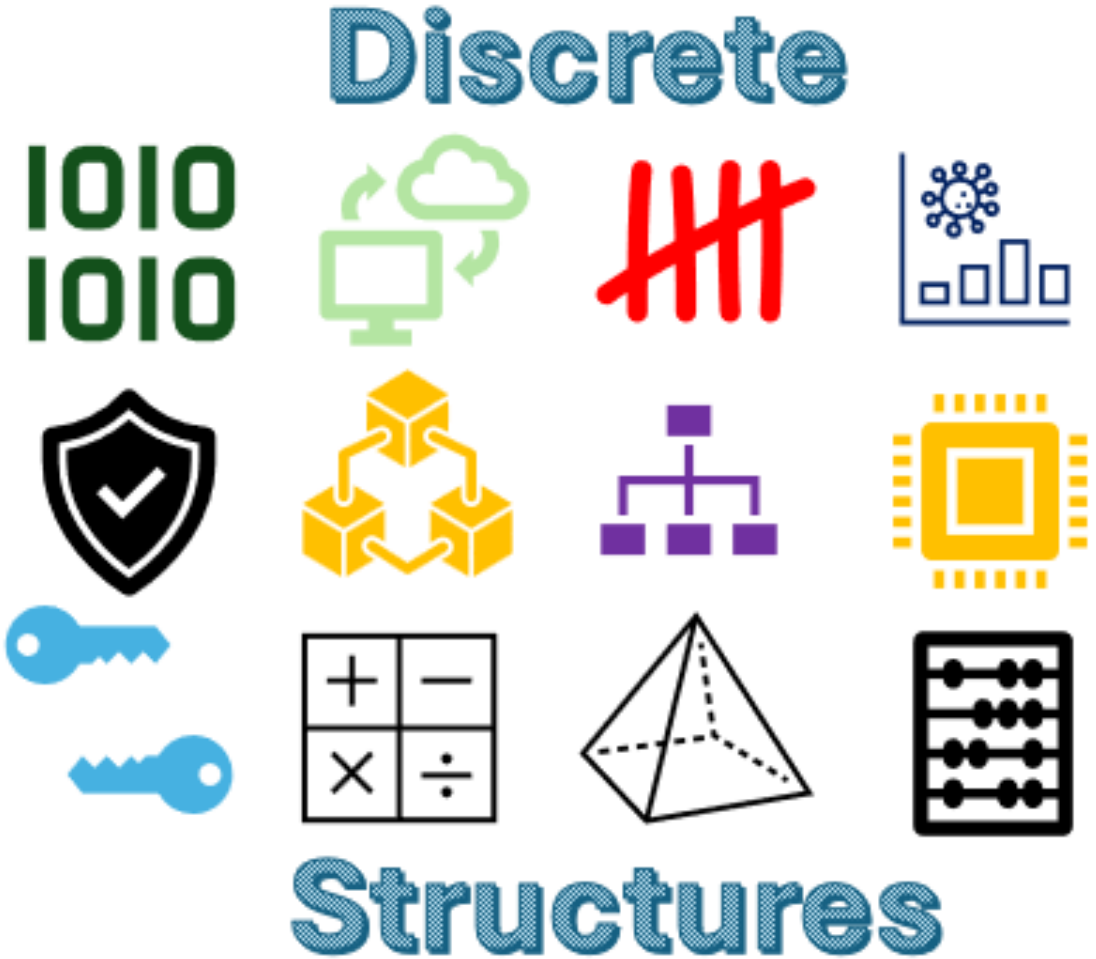
COMP 2310: Discrete Structures
Fall 2026

Chapter 1: Fundamentals



Properties of integers

Greatest Common Divisor

Euclid(m, n)
//Computes gcd(m, n) by Euclid's algorithm
//Input: Two nonnegative, not-both-zero integers m and n
//Output: Greatest common divisor of m and n
while n ≠ 0 **do**
    r ← m mod n
    m ← n
    n ← r
return m

Can you solve the problems below step-by-step?

gcd (20, 5) = ?

gcd (4, 16) = ?

Least Common Multiple (LCM)

- The smallest positive number that is a multiple of two or more given numbers.
- The LCM of two or more numbers is always greater than or equal to the largest of the numbers.

Implementation of GCD and LCM
Active class participation
Worth 1%

Matrix

Euclid(m, n)
//Computes gcd(m, n) by Euclid's algorithm
//Input: Two nonnegative, not-both-zero integers m and n
//Output: Greatest common divisor of m and n
while n ≠ 0 **do**
    r ← m mod n
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return m

Can you solve the problems below step-by-step?

gcd (20, 5) = ?

gcd (4, 16) = ?